

- [26] Lida Chen, Zujie Liang, Xintao Wang, Jiaqing Liang, Yanghua Xiao, Feng Wei, Jinglei Chen, Zhenghong Hao, Bing Han, and Wei Wang. Teaching Large Language Models to Express Knowledge Boundary from Their Own Signals. (arXiv:2406.10881), 2024.
- [27] Faeze Brahman, Sachin Kumar, Vidhisha Balachandran, Pradeep Dasigi, Valentina Pyatkin, Abhilasha Ravichander, Sarah Wiegrefe, Nouha Dziri, Khyathi Chandu, Jack Hessel, et al. The art of saying no: Contextual noncompliance in language models. *Advances in Neural Information Processing Systems*, 37: 49706–49748, 2024.
- [28] Yang Deng, Yong Zhao, Moxin Li, See-Kiong Ng, and Tat-Seng Chua. Don’t Just Say “I don’t know”! Self-aligning Large Language Models for Responding to Unknown Questions with Explanations. In Yaser Al-Onaizan, Mohit Bansal, and Yun-Nung Chen, editors, *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 13652–13673. Association for Computational Linguistics, 2024.
- [29] Bingbing Wen, Jihan Yao, Shangbin Feng, Chenjun Xu, Yulia Tsvetkov, Bill Howe, and Lucy Lu Wang. Know Your Limits: A Survey of Abstention in Large Language Models. (arXiv:2407.18418), 2025.
- [30] Felix B. Mueller, Rebekka Görges, Anna K. Bernzen, Janna C. Pirk, and Maximilian Poretschkin. Lms and memorization: On quality and specificity of copyright compliance. In Sanmay Das, Brian Patrick Green, Kush Varshney, Marianna Ganapini, and Andrea Renda, editors, *Proceedings of the Seventh AAAI/ACM Conference on AI, Ethics, and Society (AIES-24) - Full Archival Papers, October 21-23, 2024, San Jose, California, USA - Volume 1*, pages 984–996. AAAI Press, 2024. URL <https://doi.org/10.1609/aies.v7i1.31697>.
- [31] Mantas Mazeika, Long Phan, Xuwang Yin, Andy Zou, Zifan Wang, Norman Mu, Elham Sakhaee, Nathaniel Li, Steven Basart, Bo Li, David Forsyth, and Dan Hendrycks. HarmBench: A Standardized Evaluation Framework for Automated Red Teaming and Robust Refusal. (arXiv:2402.04249), 2024.
- [32] Ziwei Xu, Sanjay Jain, and Mohan Kankanhalli. Hallucination is Inevitable: An Innate Limitation of Large Language Models. (arXiv:2401.11817), 2025.
- [33] Yejin Bang, Ziwei Ji, Alan Schelten, Anthony Hartshorn, Tara Fowler, Cheng Zhang, Nicola Cancedda, and Pascale Fung. HalluLens: LLM Hallucination Benchmark. (arXiv:2504.17550), 2025.
- [34] Shangbin Feng, Weijia Shi, Yike Wang, Wenxuan Ding, Vidhisha Balachandran, and Yulia Tsvetkov. Don’t Hallucinate, Abstain: Identifying LLM Knowledge Gaps via Multi-LLM Collaboration. In Lun-Wei Ku, Andre Martins, and Vivek Srikumar, editors, *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 14664–14690. Association for Computational Linguistics, 2024.
- [35] Daniel D. Johnson, Daniel Tarlow, David Duvenaud, and Chris J. Maddison. Experts Don’t Cheat: Learning What You Don’t Know By Predicting Pairs. (arXiv:2402.08733), 2024.
- [36] Ziwei Ji, Lei Yu, Yeskendir Koishekenov, Yejin Bang, Anthony Hartshorn, Alan Schelten, Cheng Zhang, Pascale Fung, and Nicola Cancedda. Calibrating Verbal Uncertainty as a Linear Feature to Reduce Hallucinations. (arXiv:2503.14477), 2025.
- [37] Yasin Abbasi Yadkori, Ilja Kuzborskij, David Stutz, András György, Adam Fisch, Arnaud Doucet, Iuliya Beloshapka, Wei-Hung Weng, Yao-Yuan Yang, Csaba Szepesvári, Ali Taylan Cemgil, and Nenad Tomasev. Mitigating LLM Hallucinations via Conformal Abstention. (arXiv:2405.01563), 2024.
- [38] Mingda Chen, Yang Li, Karthik Padthe, Rulin Shao, Alicia Sun, Luke Zettlemoyer, Gargi Ghosh, and Wen-tau Yih. Improving Factuality with Explicit Working Memory. (arXiv:2412.18069), 2025.
- [39] Eric Zelikman, Yuhuai Wu, Jesse Mu, and Noah Goodman. Star: Bootstrapping reasoning with reasoning. *Advances in Neural Information Processing Systems*, 35:15476–15488, 2022.
- [40] Haipeng Luo, Qingfeng Sun, Can Xu, Pu Zhao, Jianguang Lou, Chongyang Tao, Xiubo Geng, Qingwei Lin, Shifeng Chen, Yansong Tang, and Dongmei Zhang. WizardMath: Empowering Mathematical Reasoning for Large Language Models via Reinforced Evol-Instruct. (arXiv:2308.09583), 2025.

- [41] Zhangyue Yin, Qiushi Sun, Qipeng Guo, Zhiyuan Zeng, Xiaonan Li, Junqi Dai, Qinyuan Cheng, Xuanjing Huang, and Xipeng Qiu. Reasoning in flux: Enhancing large language models reasoning through uncertainty-aware adaptive guidance. In Lun-Wei Ku, Andre Martins, and Vivek Srikumar, editors, *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 2401–2416. Association for Computational Linguistics, 2024. URL <https://aclanthology.org/2024.acl-long.131/>.
- [42] Aitor Lewkowycz, Anders Andreassen, David Dohan, Ethan Dyer, Henryk Michalewski, Vinay Ramasesh, Ambrose Slone, Cem Anil, Imanol Schlag, Theo Gutman-Solo, Yuhuai Wu, Behnam Neyshabur, Guy Gur-Ari, and Vedant Misra. Solving Quantitative Reasoning Problems with Language Models. *Advances in Neural Information Processing Systems*, 35:3843–3857, 2022.
- [43] An Yang, Beichen Zhang, Binyuan Hui, Bofei Gao, Bowen Yu, Chengpeng Li, Dayiheng Liu, Jianhong Tu, Jingren Zhou, Junyang Lin, Keming Lu, Mingfeng Xue, Runji Lin, Tianyu Liu, Xingzhang Ren, and Zhenru Zhang. Qwen2.5-Math Technical Report: Toward Mathematical Expert Model via Self-Improvement. (arXiv:2409.12122), 2024.
- [44] Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, Y. K. Li, Y. Wu, and Daya Guo. DeepSeekMath: Pushing the Limits of Mathematical Reasoning in Open Language Models. (arXiv:2402.03300), 2024.
- [45] Longhui Yu, Weisen Jiang, Han Shi, Jincheng Yu, Zhengying Liu, Yu Zhang, James Kwok, Zhenguo Li, Adrian Weller, and Weiyang Liu. MetaMath: Bootstrap Your Own Mathematical Questions for Large Language Models. In *The Twelfth International Conference on Learning Representations*, 2023. URL <https://openreview.net/forum?id=N8N0hgNDRt>.
- [46] Dan Hendrycks, Collin Burns, Saurav Kadavath, Akul Arora, Steven Basart, Eric Tang, Dawn Song, and Jacob Steinhardt. Measuring Mathematical Problem Solving With the MATH Dataset. (arXiv:2103.03874), 2021.
- [47] Hunter Lightman, Vineet Kosaraju, Yuri Burda, Harrison Edwards, Bowen Baker, Teddy Lee, Jan Leike, John Schulman, Ilya Sutskever, and Karl Cobbe. Let’s Verify Step by Step. In *The Twelfth International Conference on Learning Representations*, 2023. URL <https://openreview.net/forum?id=v8L0pN6EOi>.
- [48] Elliot Glazer, Ege Erdil, Tamay Besiroglu, Diego Chicharro, Evan Chen, Alex Gunning, Caroline Falkman Olsson, Jean-Stanislas Denain, Anson Ho, Emily de Oliveira Santos, Olli Järvinen, Matthew Barnett, Robert Sandler, Matej Vrzala, Jaime Sevilla, Qiuyu Ren, Elizabeth Pratt, Lionel Levine, Grant Barkley, Natalie Stewart, Bogdan Grechuk, Tetiana Grechuk, Shreepranav Varma Enugandla, and Mark Wildon. FrontierMath: A Benchmark for Evaluating Advanced Mathematical Reasoning in AI. (arXiv:2411.04872), 2024.
- [49] Haoxiang Sun, Yingqian Min, Zhipeng Chen, Wayne Xin Zhao, Lei Fang, Zheng Liu, Zhongyuan Wang, and Ji-Rong Wen. Challenging the Boundaries of Reasoning: An Olympiad-Level Math Benchmark for Large Language Models. (arXiv:2503.21380), 2025.
- [50] Mislav Balunović, Jasper Dekoninck, Ivo Petrov, Nikola Jovanović, and Martin Vechev. Matharena: Evaluating llms on uncontaminated math competitions. 2025. URL <https://matharena.ai/>.
- [51] Jingyuan Ma, Damai Dai, Zihang Yuan, Rui li, Weilin Luo, Bin Wang, Qun Liu, Lei Sha, and Zhifang Sui. Large Language Models Struggle with Unreasonability in Math Problems. (arXiv:2403.19346), 2025.
- [52] A. M. Muntasir Rahman, Junyi Ye, Wei Yao, Sierra S. Liu, Jesse Yu, Jonathan Yu, Wenpeng Yin, and Guiling Wang. From Blind Solvers to Logical Thinkers: Benchmarking LLMs’ Logical Integrity on Faulty Mathematical Problems. (arXiv:2410.18921), 2025.
- [53] Freda Shi, Xinyun Chen, Kanishka Misra, Nathan Scales, David Dohan, Ed H. Chi, Nathanael Schärli, and Denny Zhou. Large Language Models Can Be Easily Distracted by Irrelevant Context. In *Proceedings of the 40th International Conference on Machine Learning*, pages 31210–31227. PMLR, 2023.
- [54] Jialin Ouyang. TreeCut: A Synthetic Unanswerable Math Word Problem Dataset for LLM Hallucination Evaluation. (arXiv:2502.13442), 2025.

- [55] Zihao Zhou, Shudong Liu, Maizhen Ning, Wei Liu, Jindong Wang, Derek F. Wong, Xiaowei Huang, Qiufeng Wang, and Kaizhu Huang. Is Your Model Really A Good Math Reasoner? Evaluating Mathematical Reasoning with Checklist. (arXiv:2407.08733), 2024.
- [56] Asir Saadat, Tasmia Binte Sogir, Md Taukir Azam Chowdhury, and Syem Aziz. When Not to Answer: Evaluating Prompts on GPT Models for Effective Abstention in Unanswerable Math Word Problems. (arXiv:2410.13029), 2024.
- [57] YuHong Sun, Zhangyue Yin, Qipeng Guo, Jiawen Wu, Xipeng Qiu, and Hui Zhao. Benchmarking Hallucination in Large Language Models Based on Unanswerable Math Word Problem. In Nicoletta Calzolari, Min-Yen Kan, Veronique Hoste, Alessandro Lenci, Sakriani Sakti, and Nianwen Xue, editors, *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*, pages 2178–2188. ELRA and ICCL, 2024. URL <https://aclanthology.org/2024.lrec-main.196/>.
- [58] Xunjian Yin, Baizhou Huang, and Xiaojun Wan. ALCUNA: Large Language Models Meet New Knowledge. In Houda Bouamor, Juan Pino, and Kalika Bali, editors, *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 1397–1414. Association for Computational Linguistics, 2023.
- [59] Alicia Parrish, Angelica Chen, Nikita Nangia, Vishakh Padmakumar, Jason Phang, Jana Thompson, Phu Mon Htut, and Samuel Bowman. BBQ: A hand-built bias benchmark for question answering. In Smaranda Muresan, Preslav Nakov, and Aline Villavicencio, editors, *Findings of the Association for Computational Linguistics: ACL 2022*, pages 2086–2105. Association for Computational Linguistics, 2022.
- [60] Aarohi Srivastava, Abhinav Rastogi, Abhishek Rao, . . . , Zijie J. Wang, Zirui Wang, and Ziyi Wu. Beyond the imitation game: Quantifying and extrapolating the capabilities of language models. *Transactions on Machine Learning Research*, 2023. ISSN 2835-8856. URL <https://openreview.net/forum?id=uyTL5Bvosj>.
- [61] Shengding Hu, Yifan Luo, Huadong Wang, Xingyi Cheng, Zhiyuan Liu, and Maosong Sun. Won’t Get Fooled Again: Answering Questions with False Premises. In Anna Rogers, Jordan Boyd-Graber, and Naoaki Okazaki, editors, *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 5626–5643. Association for Computational Linguistics, 2023.
- [62] Tu Vu, Mohit Iyyer, Xuezhi Wang, Noah Constant, Jerry Wei, Jason Wei, Chris Tar, Yun-Hsuan Sung, Denny Zhou, Quoc Le, and Thang Luong. FreshLLMs: Refreshing Large Language Models with Search Engine Augmentation. (arXiv:2310.03214), 2023.
- [63] Shuyue S. Li, Vidhisha Balachandran, Shangbin Feng, Jonathan S. Ilgen, Emma Pierson, Pang W. Koh, and Yulia Tsvetkov. MediQ: Question-Asking LLMs and a Benchmark for Reliable Interactive Clinical Reasoning. *Advances in Neural Information Processing Systems*, 37:28858–28888, 2024.
- [64] Nino Scherrer, Claudia Shi, Amir Feder, and David Blei. Evaluating the Moral Beliefs Encoded in LLMs. *Advances in Neural Information Processing Systems*, 36:51778–51809, 2023.
- [65] Najoung Kim, Phu Mon Htut, Samuel R. Bowman, and Jackson Petty. (QA)²: Question Answering with Questionable Assumptions. In Anna Rogers, Jordan Boyd-Graber, and Naoaki Okazaki, editors, *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 8466–8487. Association for Computational Linguistics, 2023.
- [66] Pradeep Dasigi, Kyle Lo, Iz Beltagy, Arman Cohan, Noah A. Smith, and Matt Gardner. A Dataset of Information-Seeking Questions and Answers Anchored in Research Papers. In Kristina Toutanova, Anna Rumshisky, Luke Zettlemoyer, Dilek Hakkani-Tur, Iz Beltagy, Steven Bethard, Ryan Cotterell, Tanmoy Chakraborty, and Yichao Zhou, editors, *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 4599–4610. Association for Computational Linguistics, 2021.
- [67] Michael Zhang and Eunsol Choi. SituatedQA: Incorporating Extra-Linguistic Contexts into QA. In Marie-Francine Moens, Xuanjing Huang, Lucia Specia, and Scott Wen-tau Yih, editors, *Proceedings of*